

Teo central limite

Teorema 1 (central del límite). Sean $(X_j)_{j \in \mathbb{N}} \subset \mathcal{L}^2(\mathbb{P})$ v.a.i.i.d.

$$\implies \frac{S_n - n \cdot \mathbb{E}[X_1]}{\sqrt{n} \sqrt{\mathbb{V}(X_1)}} \xrightarrow[n \rightarrow \infty]{d} Z \quad \text{con} \quad \forall t \in \mathbb{R} : f_Z(t) = \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}.$$